MILITARY SPECIFICATION

PLUG ASSEMBLY, LIFTING, UNIVERSAL

This specification is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, provides quality assurance provisions, and packaging for the metal parts and assembly for one type of lifting plug designated as Universal Plug Assembly for 155MM and 8 Inch Projectiles.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and Standards. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
PPP-B-587	-	Box, Wood, Wirebound Pallet Type

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9345325 - Plug, Universal, Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adpoted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 950-54 - Standard Method of Test for Impact Strength of Adhesive Bonds

(Applications for copies should be addressed to: American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies, shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Tension. The plug shall sustain a load of 8,000 $\pm$ 50 pounds without any sign of failure.

3.3 Ductility. The plug shall withstand deformation of the eye without cracking.

3.4 Metal imperfections. The plug shall be free from porosity shrinks or discontinuous structure.

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3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Threads shall be complete and undamaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Universal Lifting Plug Assembly	9345325	50
Universal Plug	9347368	25 min. (at least 3 per cavity)
Fusible Alloy	9215391	1 lb.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Plug, Assembly, Lifting, Universal				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Universal Plug Assembly (Dwg. 9345325)				
	Examination for defects	50	-	3.2	4.4.2.3
	Air pressure test	50	-	3.2	4.5.2
	Universal Plug (Dwg. 9347368)	Note a			
	Examination for defects		-	3.2	4.4.2.1
	Fusible Alloy (Dwg. 9215391)				
	Melting test	1 lb	-	3.2	4.5.1

NOTES:
a. Twenty-five (25) minimum; at least 3 per cavity.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Ternary Eutectic Alloy from one lot number.

b. Plugs of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Dimensional control of cast parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least five (5) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of every 5000 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

Use of these plans are limited to those dimensions and features controlled by the mold cavity. For all specified tests, visual examinations and examinations of dimensions or features not developed exclusively by the mold cavity, these optional plans are not applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9347368
4.4.2.1	Plug, Universal				NEXT HIGHER ASSEMBLY 9345325
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major					
101	Width of eye section, min.		0.40%	3.1	Gage
102	Length of thread section		0.40%	3.1	Gage
103	Dimension from bottom of thread section to bottom of flange		0.40%	3.1	Gage
104	Outside diameter of shank		0.40%	3.1	Gage
105	Flange thickness		0.40%	3.1	Gage
106	Diameter of flange, min.		0.40%	3.1	Gage
107	Diamenter of gasket groove		0.40%	3.1	Gage
108	Depth of gasket groove, max.		0.40%	3.1	Gage
109	True position of gasket groove		0.40%	3.1	Gage
110	Perpendicularity of flange		0.40%	3.1	Gage
111	Thread pitch diameter, min.		0.40%	3.1	Gage
112	Thread major diameter, min.		0.40%	3.1	Gage
113	Tension test (see Note a)	5 min		3.2	4.5.3
114	Ductility test (see Note b)	5 min		3.3	4.5.4
115	Metal imperfection (see Note a)	5 min		3.4	4.5.5
116	Radius at base of shank, max.		0.40%	3.1	Gage
117	Inside diameter		0.40%	3.1	Gage
118	True position of shank		0.40%	3.1	Gage
119	True position of thread Diameter		0.40%	3.1	Gage

NOTES:
a. At least one per cavity.
b. Same samples as Major 113.

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9347368
4.4.2.1	Plug, Universal				NEXT HIGHER ASSEMBLY 9345325
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Thread pitch diameter, max.		0.65%	3.1	Gage
202	Width of gasket groove		0.65%	3.1	Gage
203	True position of flange		0.65%	3.1	Gage
204	Radius at top of shank, min.		0.65%	3.1	Gage
205	Outside radius of eye, min.		0.65%	3.1	Gage
206	Height of eye hole		0.65%	3.1	Gage
207	Width of eye hole		0.65%	3.1	Gage
208	Angle of groove inside diameter		0.65%	3.1	Gage
209	Cavity identification missing		0.65%	3.1	Visual
210	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
211	Surface finish incorrect		0.65%	3.1	Visual
212	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9347368
4.4.2.2	Universal Plug Assembly (Prior to Pouring Alloy)				NEXT HIGHER ASSEMBLY 9345325
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major 101	Primer or sealing compound not present on noted surface		0.40%	3.1	Visual
Minor:	None defined.				

NOTES:

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DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345325
4.4.2.3	Universal Plug Assembly (After Pouring Alloy)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major		Note a			
101	Salt spray			3.1	4.5.7
102	Air pressure test		100%	3.1	4.5.2
103	Alloy excessively below or excessively above noted surface		0.40%	3.1	Gage
Minor					
201	Paint missing, incomplete or on undersurface		0.65%	3.1	Visual
202	Anti-seize compound missing or incomplete		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

a. Sampling plan as specified in TT-C-490.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215391
4.4.2.4	Ternary Eutectic Alloy				NEXT HIGHER ASSEMBLY 9345325
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major 101	Melting test	5	-	3.1	4.5.1
Minor:	None defined.				

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70438 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215393
4.4.2.5	Adhesive				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined.				
Major 101	Impact resistance test	10	-	3.1	4.5.6
Minor:	None defined.				

NOTES:

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4.4.3 Testing. The tests (First Article and Quality Conformance) listed in the Item Detail Specification shall be performed on inspection lots as defined in 4.4.1 of this specification.

4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Methods of inspection.

4.5.1 Melting test of ternary eutectic alloy. Five (5) samples of approximately one quarter (1/4) pound each shall be randomly taken from each lot of ternary eutectic alloy to be used in each lot of fusible plug assemblies. The samples selected shall be individually tested by placing each in a beaker of distilled water at ambient temperature. The temperature of the water shall then be raised at a rate not to exceed $10^{\circ}F$ per minute. The water shall be stirred to assure uniform heating. A temperature-indicating device such as a thermometer or thermocouple, suitable for the required temperature range, shall be inserted in the water. When the alloy is observed to melt, the indicated temperature shall be recorded. The measured temperature shall then be compared with the value specified on the applicable drawing to determine compliance or non-compliance of the sample with the melting point requirement. If any sample fails to comply with the specified melting point, the alloy lot shall be rejected. In addition, all fusible plug assemblies containing the rejected lot of ternary eutectic alloy shall be rejected.

4.5.2 Air pressure test of fusible plug assembly. Every plug shall be subjected to this test. The test shall be performed by sealing the open end of the plug and completely immersing the plug in water. The air pressure specified on the applicable drawing shall then be applied through the sealed end of the plug for the specified time intended. Observation shall be made for air bubbles originating from the area of the fusible alloy indicating a leaking plug. All such leaking plugs shall be classed defective and removed from the lot. The plug assemblies shall be thoroughly dried after completion of this test.

All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated

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prior to the start of each shift. At the start of each shift and every four (4) hours thereafter, the equipment operation shall be verified with a known leaker. If the equipment fails either to calibrate or verify, all plugs tested since the last satisfactory check will be retested.

4.5.3 Tension. Five (5) lifting plugs shall be selected from each lot after final heat treatment and prior to pouring of the alloy for this test. The lifing plug shall be screwed into a threaded plate or other suitable fixture and a load of 8,000 ± 50 pounds applied through the medium of a 3/4 inch diameter bolt or pin passing through the eye of the plug. The full load shall be transmitted through the screw threads of the plug. Visual observation shall be made to determine compliance with 3.2. If any plug fails to comply with the requirement, the lot represented by the plugs shall be rejected.

4.5.4 Ductility. The five (5) samples from the tension test shall be used for this test. The plugs shall be cooled to minus 40 degrees Fahrenheit (F) for a minimum of thirty (30) minutes. The eye of each plug shall be flattened by either pressure or blows in any direction until the inner diameter of the eye is reduced by 0.050 inch while at minus 40 degrees F. Visual observation shall be made for any evidence of cracking. If any plug fails to meet the applicable requirement, the lot represented by these samples shall be rejected.

4.5.5 Metal imperfections. Five (5) lifting plugs shall be selected from each heat treatment lot after final heat treatment for this test. The body of each sample malleable iron casting shall be cut through the thickest section for inspection. The plug shall be observed for discontinuous structure, porosity or shrink. If any sample fails to comply with the requirement, the batch represented by the samples shall be rejected.

4.5.6 Impact resistance of adhesive. The bottle of adhesive representing the lot which will be used for a week's production will be tested prior to use for impact resistance. A total of ten specimens shall be prepared and tested in accordance with the procedures and methods specified in ASTM D-950-54. If any sample tested fails to meet an impact resistance of ten minus one (10-1) ft. lbs. (foot pounds), the lot or batch of adhesive shall be rejected.

4.5.7 Salt spray test. The sampling plan for this test shall be as specified in TT-C-490. The test shall be performed on the plug assemblies as specified in TT-C-490. The number of hours and the percentage of sodium chloride solution shall be as indicated on

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the applicable drawing. Examination shall be made for any evidence of corrosion of the painted areas. If any plug assembly fails to comply with the specified requirements, it shall be classified defective and corrective actions taken as specified in TT-C-490.

5. PACKAGING

5.1 Packing and marking.

5.1.1 Overseas shipment. Lifting plugs shall be semi-bulk packaged, moisture-proof packaged (barrier bag only), packed, marked and palletized in accordance with MIL-STD-1169, Level A. Plugs shall be separated in the unit pack to protect the paint surfaces.

5.1.2 Domestic shipment. Lifting plugs shall be semi-bulk packaged, packed and marked in accordance with MIL-STD-1169, Level B. Palletization is optional. Plugs shall be separated in the unit packaged to protect the paint surfaces.

5.1.2.1 Alternate pack. As an alternate pack to the above, lifting plugs may be semi-bulk packed in a Type I, Class 2 pack per PPP-B-587. Plugs shall be separated in the box to protect paint surfaces. The pallet shall be modified as necessary to assure that the load arrives at its destination intact, with protected cargo and in suitable condition for handling, storage and intended use.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use in lifting and handling artillery projectiles.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Center (ARDC), ATTN: AMSMC-QAR-I(D), Dover, NJ 07801.

6.4 Data requirements. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

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6.6 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery.

However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Visual examination qualification. When compliance with applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

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